

FILTERIZATION

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ABSTRACT. x

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1. VF-CATEGORIES

1.1. Blass’ category of filters.

Definition 1.1. The *category of filters* $\mathbb{F}$ is defined as follows:

- as objects, we let pairs (S, σ) of a set S and a filter σ on S ;
- as morphisms $(T, \tau) \rightarrow (S, \sigma)$, we let equivalence classes of functions $\phi: T \rightarrow S$ such that $\phi^{-1}(S_0) \in \tau$ for each $S_0 \in \sigma$, i.e. $\sigma \subseteq F\phi(\tau)$ where $F: \mathbf{Set} \rightarrow \mathbf{Pos}$ is the (relative) filter monad, identified up to equality on a τ -large subset.

For ease of notation, we will often omit the set underlying a filter.

Recall that an arrow $\phi: (T, \tau) \rightarrow (S, \sigma)$ in $\mathbb{F}$ is monic if and only if it is injective on a τ -large subset, in which case it can be identified with the inclusion of a σ -large subset; on the other hand, it is epic if and only if $\phi(T_0) \in \sigma$ for each $T_0 \in \tau$, or equivalently $\sigma = F\phi(\tau)$, in which case it can be assumed to be surjective. Moreover, $\mathbb{F}$ is balanced.

Note also that, for a set S , the hom-set $\mathbb{F}(\tau, (S, \{S\}))$ is in bijection with the *reduced power* of S with respect to τ , generalizing ultrapowers to filters.

We note here the following property of $\mathbb{F}$, first remarked by the authors of [GMT26].

Lemma 1.2. *For any epimorphism $f: \mu \twoheadrightarrow \nu$ in $\mathbb{F}$, there exist a set K , a filter κ on K , and an arrow $g: \kappa \otimes \nu \rightarrow \mu$ such that $f \circ g$ coincides with the projection $\pi_J: \kappa \otimes \nu \rightarrow \nu$:*

$$\begin{array}{ccc}
 & & \mu \\
 & \nearrow g & \downarrow f \\
 \kappa \otimes \nu & \xrightarrow{\pi_J} & \nu
 \end{array}$$

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Proof. Up to isomorphism in $\mathbb{F}$, we may pick a surjective representative $f: I \twoheadrightarrow J$ for the arrow $f: (I, \mu) \rightarrow (J, \nu)$. Let K be the set $\{k: J \rightarrow I \mid fk = \text{id}_J\}$ of sections of f , which is non-empty as f is surjective, and define $g: K \times J \rightarrow I$ by $g(k, j) := k(j)$, which clearly gives $f \circ g = \pi_J$.

For each $A \in \mu$, consider the set $N_A := \{k \in K \mid (kf)^{-1}(A) \in \mu\}$. We claim that the family $\{N_A \mid A \in \mu\}$ has the finite intersection property. As $N_A \cap N_B = N_{A \cap B}$ for all $A, B \in \mu$, it suffices to prove $N_A \neq \emptyset$ for each $A \in \mu$. Indeed, given $A \in \mu$, define $k \in K$ by setting $k(j)$ to be any element of $A \cap f^{-1}(j)$ if $j \in f[A]$, and an arbitrary element of $f^{-1}(j)$ otherwise. Then $A \subseteq (kf)^{-1}(A)$, so that $(kf)^{-1}(A) \in \mu$ and hence $k \in N_A$. **UT:** this uses classical logic!! (but not choice)

Consider then the filter κ on K generated by $\{N_A \mid A \in \mu\}$. We claim that $\mu \subseteq g(\kappa \otimes \nu)$, so that g defines an arrow $g: \kappa \otimes \nu \rightarrow \mu$ in $\mathbb{F}$. Let $A \in \mu$. Note that $\sum_{k \in N_A} k^{-1}(A) \subseteq g^{-1}(A)$, by definition of N_A . Now, for each $k \in N_A$, we have $k^{-1}(A) \in f(\mu) = \nu$, and by construction $N_A \in \kappa$. Thus, $\sum_{k \in N_A} k^{-1}(A) \in \kappa \otimes \nu$, so that $g^{-1}(A) \in \kappa \otimes \nu$ as well, i.e., $A \in g(\kappa \otimes \nu)$. $\square$

GS: What we need to make it work is that $f: \tau \twoheadrightarrow \sigma$ satisfies that $\int_{s:\sigma} T^0_s$ is non-empty for all $T^0 \subseteq T$ that is ν -large. Maybe we can replace, in the statement, epi by map satisfying this property?

Lemma 1.3.

1.2. Vf-categories.

Definition 1.4. A *vf-category* $\mathcal{X}$ is defined by the following data:

- (i) a class $\mathcal{X}_0$, whose elements are called *points*;
- (ii) for each $p \in X_0$ and each $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$ for some filter $\sigma \in \mathbb{F}$, a functor:

$$\mathcal{X}_{p,(q_s)}: (\mathbb{F}/\sigma)^{\text{op}} \rightarrow \mathbf{Set} \quad (\phi: \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t:\tau})$$

where elements of $\mathcal{X}(p, (q_{\phi t})_{t:\tau})$ are called *vf-arrows* and denoted by $r: p \multimap_{t:\tau} q_{\phi t}$ or simply $r: p \multimap_\tau q_{\phi t}$;

- (iii) for each $p, (q_s)_{s:\sigma}, (r_{s,t})_{(s,t):\sum_{s:\sigma} \tau_s}$, a *unit* and a *composition*:

$$\begin{array}{ccccc} 1 & \longrightarrow & 1 & (\mathbb{F}/\sigma \times \int_{s:\sigma} (\mathbb{F}/\tau_s))^{\text{op}} & \xrightarrow{\mathcal{X}_{p,(q_{-t})} \times \int_{s:\sigma} \mathcal{X}_{q_s,(r_{s,-k})}} & \mathbf{Set} \times \mathbf{Set} \\ \star \downarrow & & \Downarrow & \Sigma \downarrow & & \Downarrow \\ \mathbb{F}^{\text{op}} & \xrightarrow{H_{p,(p)}} & \mathbf{Set} & (\mathbb{F}/\sum_{s:\sigma} \tau_s)^{\text{op}} & \xrightarrow{H_{p,-}} & \mathbf{Set} \end{array}$$

This data is required to satisfy the following axioms:

- (i) **axioms**

We say that $\mathcal{X}$ is *small* if so is X_0 .

For any vf-arrow $r: p \multimap_{s:\sigma} q_s$ and any $\phi: \tau \rightarrow \sigma$ in $\mathbb{F}/\sigma$ — seen as an arrow $\phi \rightarrow \text{id}_\sigma$ in $\mathbb{F}/\sigma$ — we denote by $\phi^*(r)$ the vf-arrow $\mathcal{X}_{p,(q_s)}(\phi)(r): p \multimap_{t:\tau} q_{\phi t}$, which we refer to as the *reindexing* of r along ϕ .

- If ϕ is monic, we also say that $\phi^*(r)$ is a *specialization* of r .
- If ϕ is epic, we also say that $\phi^*(r)$ is a *thickening* of r .

1.3. Tautness. Let $\mathcal{X}$ be a vf-category. Note first that if a vf-arrow $u: p \multimap_\tau q_{\phi t}$ is the thickening of $v: p \multimap_\sigma q_s$ with respect to some $\mathbb{F}$ -arrow $\phi: \tau \twoheadrightarrow \sigma$, i.e. $u = \phi^*v$, then in particular $\psi_1^*u = \psi_2^*v$ for any pair of $\mathbb{F}$ -arrows ψ_1, ψ_2 such that $\phi \circ \psi_1 = \phi \circ \psi_2$.

Definition 1.5. A vf-arrow $u: p \multimap_{\tau} q_{\phi t}$ in $\mathcal{X}$ is ϕ -reducible (or reducible along ϕ) if $\psi_1^* u = \psi_2^* u$ for every pair of parallel $\mathbb{F}$ -arrows ψ_1, ψ_2 such that $\phi \circ \psi_1 = \phi \circ \psi_2$.

As we now define, in a *taut* vf-category, reducible vf-arrows can be uniquely reduced.

Definition 1.6. A vf-category $\mathcal{X}$ is *taut* if:

- (i) for every ϕ -reducible vf-arrow $u: p \multimap_{\tau} q_{\phi t}$ there exists a unique vf-arrow $u_0: p \multimap_{\sigma} q_s$ such that $u = \phi^* u_0$;
- (ii) the canonical map $\mathcal{X}(p, (q_t)_{t: \coprod_i \sigma_i}) \rightarrow \prod_i \mathcal{X}(p, (q_s)_{s: \sigma_i})$ defined by specialization along each coproduct injection $\sigma_i \hookrightarrow \coprod_i \sigma_i$ is a bijection.

Example 1.7. Tautness can be defined analogously for vu-categories, disregarding the second condition; a vu-poset is taut if and only if it is a topological space. **UT: for vf-posets, this is *stronger* than the the condition characterizing relational F -algebras**

Construction 1.8. For a vf-poset X , its *tautification* is the vf-poset $\rho_T(X)$ defined as follows:

- (i) points of $\rho_T(X)$ are the same as those of X ;
- (ii) for each $p \in X_0$ and $(q_s)_{s: \sigma} \in X_0^{\sigma}$, $p \preceq_{\sigma} q_s$ holds if and only if there exists a cover $\phi: \tau \twoheadrightarrow \sigma$ in $\mathbb{F}$ such that $p \preceq_{\tau} q_{\phi t}$ holds in X .

It is evident that $\rho_T(X)$ is taut, and there is a canonical vf-functor $\eta_X: X \rightarrow \rho_T(X)$ acting as the identity on points.

We can now give two equivalent characterizations of tautness.

First, recall from [AE04] that we can consider a Grothendieck topology J_{∞} on the category $\mathbb{F}$ of filters, where coverings of $\sigma \in \mathbb{F}$ are generated by (possibly infinite) jointly-epimorphic families, i.e. families $\{\gamma_i: \sigma_i \rightarrow \sigma\}_{i \in I}$ such that $\bigcup_{i \in I} \gamma_i(S_i) \in \sigma$ for any $\{S_i \in \sigma_i\}_{i \in I}$. We also denote by J_{∞} the topology induced on each slice category of $\mathbb{F}$, i.e. where a covering of $\phi: \tau \rightarrow \sigma$ in $\mathbb{F}/\sigma$ is a covering of τ in $\mathbb{F}$ seen as a collection of morphisms to ϕ in $\mathbb{F}/\sigma$.

Theorem 1.9. Let $\mathcal{X}$ be a vf-category. The following are equivalent:

- (i) $\mathcal{X}$ is taut;
- (ii) for any small vf-poset Y and any vf-functor $F: Y \rightarrow \mathcal{X}$, there exists a unique lift $\bar{F}: \rho_T(Y) \rightarrow \mathcal{X}$ of F along the tautification map $\eta_Y: Y \rightarrow \rho_T(Y)$:

$$\begin{array}{ccc} Y & \xrightarrow{F} & \mathcal{X} \\ \eta_Y \downarrow & \nearrow \bar{F} & \\ \rho_T(Y) & & \end{array}$$

- (iii) for any $p \in \mathcal{X}_0$ and any $(q_s)_{s: \sigma} \in \mathcal{X}_0^{\sigma}$, the functor $\mathcal{X}_{p, (q_s)}: (\mathbb{F}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$ mapping $(\phi: \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t: \tau})$ is a J_{∞} -sheaf on $\mathbb{F}/\sigma$.

Proof. Towards (1) $\Rightarrow$ (3), consider a (generating) J_{∞} -covering family $\{\gamma_i: \tau_i \rightarrow \tau\}_{i \in I}$ of $\phi: \tau \rightarrow \sigma$ in $\mathbb{F}/\sigma$ and a compatible family $\{u_i: p \multimap_{\tau_i} q_{\phi \gamma_{it}}\}_{i \in I}$ in $\mathcal{X}_{p, (q_s)}$. We need to find a unique $u_0: p \multimap_{\tau} q_{\phi t}$ such that $u_i = \gamma_i^* u_0$ for each $i \in I$.

Consider the coproduct map $\gamma: \coprod_i \tau_i \twoheadrightarrow \tau$ and denote by $\iota_i: \tau_i \rightarrow \coprod_i \tau_i$ the coproduct injection at each $i \in I$. By tautness, the family $\{u_i: p \multimap_{\tau_i} q_{\phi \gamma_{it}}\}_{i \in I}$ uniquely detects a vf-arrow $u: p \multimap_{\coprod_i \tau_i} q_{\phi \gamma t}$ such that $u_i = \iota_i^* u$ for each $i \in I$. Note now that u is γ -reducible, since coproducts in $\mathbb{F}$ are disjoint **UT: are they?** and $\{u_i\}_{i \in I}$ is a compatible family in $\mathcal{X}_{p, (q_s)}$. Therefore, again by tautness it follows that there exists a unique vf-arrow $u_0: p \multimap_{\tau} q_{\phi t}$ such that $u = \gamma^* u_0$; this vf-arrow then clearly satisfies $u_i = \gamma_i^* u_0$ for each $i \in I$. To show uniqueness, suppose that v_0 is another vf-arrow satisfying $u_i = \gamma_i^* v_0$ for

each $i \in I$; then, $\iota_i^* u = u_i = \gamma_i^* v_0$ for each $i \in I$, so that necessarily $\gamma^* v_0 = u$ by the uniqueness property of u and hence $v_0 = u_0$ by the uniqueness property of u_0 .

(3) $\Rightarrow$ (2)

(2) $\Rightarrow$ (1)

□

Remark 1.10. The Grothendieck topology J_∞ restricts to the atomic topology on $\mathbb{U}$. we expect that [vu-categories and vf-categories are equivalent notions \(see Vu-vF appendix\)](#)

1.4. Fully-faithfulness.

Definition 1.11. A vf-functor $F: \mathcal{X} \rightarrow \mathcal{Y}$ is *vf-faithful* (resp. *vf-fully-faithful*) if, for each $p \in \mathcal{X}_0$ and each $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$, the function

$$\mathcal{X}(p, (q_s)_{s:\sigma}) \rightarrow \mathcal{Y}(F(p), (F(q_s))_{s:\sigma})$$

is injective (resp. bijective).

Definition 1.12. A vf-functor $F: \mathcal{X} \rightarrow \mathcal{Y}$ is *vf-full* if for each vf-arrow $u: F(x) \multimap_{s:\sigma} F(y_s)$ in $\mathcal{Y}$ there exist a covering $\{\phi_i: \tau_i \rightarrow \sigma\}_{i \in I}$ of σ in $\mathbb{F}$ and a family of vf-arrows $\{v_i: x \multimap_{t:\tau_i} y_{\phi_{it}}\}_{i \in I}$ in $\mathcal{X}$ such that $F(v_i) = \phi_i^*(u)$ for each $i \in I$.

Remark 1.13. Taking coproducts in $\mathbb{F}$, the previous definition can be reduced to the case $I = 1$; moreover, by Lemma 1.2 we can assume $\phi = \pi_2: \lambda \cdot \sigma \rightarrow \sigma$, i.e., there exists an arrow $g: p \multimap_{\lambda \cdot \sigma} ((q_s)_{s:\sigma})_{\ell:\lambda}$, with λ inhabited, such that $F(g) = \pi_2^*(f)$. [GS: a filter \$\sigma\$ is inhabited if \$\sigma \rightarrow \star\$ is a covering, classically it means that it is not the initial filter UT: a priori Lemma 1.2 uses classical logic so maybe we avoid it if possible/if we can't make it constructive](#)

Lemma 1.14. Let $F: \mathcal{X} \rightarrow \mathcal{Y}$ be a vf-functor and suppose that $\mathcal{X}$ and $\mathcal{Y}$ are taut. Then, the following are equivalent:

- (i) F is vf-full and vf-faithful;
- (ii) F is vf-fully-faithful.

Proof. Clearly, (2) $\Rightarrow$ (1). Conversely, suppose F is vf-full and vf-faithful. To show that it is vf-fully-faithful we need to show that, for each $p \in \mathcal{X}_0$ and each $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$, the function

$$\mathcal{X}(p, (q_s)_{s:\sigma}) \rightarrow \mathcal{Y}(F(p), (F(q_s))_{s:\sigma})$$

is surjective, so let $u: F(p) \multimap_\sigma F(q_s)$ be a vf-arrow in $\mathcal{Y}$. By vf-fullness of F , there exist a cover $\phi: \tau \twoheadrightarrow \sigma$ and a vf-arrow $v: p \multimap_\tau q_{\phi t}$ in $\mathcal{X}$ such that $F(v) = \phi^* u$. Now, for any $\psi_1, \psi_2: \lambda \rightarrow \tau$ equalizing ϕ , we have

$$F(\psi_1^* v) = \psi_1^* F(v) = \psi_1^* \phi^* u = \psi_2^* \phi^* u = \psi_2^* F(v) = F(\psi_2^* v)$$

and hence $\psi_1^* v = \psi_2^* v$ by vf-faithfulness of F . In other words, v is unthickenable: since $\mathcal{X}$ is taut, it follows that there is a (unique) vf-arrow $v_0: p \multimap_\sigma q_s$ such that $v = \phi^* v_0$. Thus,

$$\phi^* u = F(v) = F(\phi^* v_0) = \phi^* F(v_0)$$

and hence $u = F(v_0)$ by tautness of $\mathcal{Y}$. □

1.5. Subspaces. Let $\mathcal{X}$ be a vf-category. For $p \in \mathcal{X}_0$ and $(q_s)_{s:\sigma} \in \mathcal{X}_0^\sigma$, we write $p \preceq (q_s)_{s:\sigma}$ if the set $\mathcal{X}(p, (q_s)_{s:\sigma})$ is inhabited.

Definition 1.15. A *subspace* $\mathcal{Y}$ of $\mathcal{X}$ is simply a sub-vf-category, i.e. a vf-category defined by restricting the functors $\mathcal{X}_{p,(q_s)}$ to a subclass $\mathcal{Y}_0$ of $\mathcal{X}_0$, together with the same units and compositions. Clearly, the inclusion $\mathcal{Y}_0 \hookrightarrow \mathcal{X}_0$ extends to a vf-fully-faithful vf-functor $\mathcal{Y} \rightarrow \mathcal{X}$.

A subspace $\mathcal{Y} \subseteq \mathcal{X}$ is *open* if, for any $p \in \mathcal{B}_0$ and any $(q_s)_{i:\sigma} \in \mathcal{X}_0^\sigma$, if $p \preceq (q_s)_{s:\sigma}$ then $(\forall s : \sigma) q_s \in \mathcal{B}_0$. **GS:** Open subspaces of $\mathcal{X}$ correspond exactly to a vf-functor $\mathcal{X} \rightarrow [2]$ into the Sierpinski vf-category.

If $\mathcal{X}$ is small, then the poset of open subspaces of $\mathcal{X}$ ordered by inclusion is a frame, which we denote by $\mathcal{O}(\mathcal{X})$. Note that every vf-functor $\mathcal{X} \rightarrow \mathcal{Y}$ induces, by precomposition, a frame homomorphism $\mathcal{O}(\mathcal{Y}) \rightarrow \mathcal{O}(\mathcal{X})$: this association determines a 2-functor $\mathbf{vfCat} \rightarrow \mathbf{Loc}$ and hence a 2-functor $\rho_T : \mathbf{vfCat} \rightarrow \mathbf{TopSp}$, the *topological reflection*, which is left 2-adjoint to the embedding $\iota_T : \mathbf{TopSp} \hookrightarrow \mathbf{vfCat}$.

1.6. Vf-categories of points. Let $\mathcal{E}$ be a topos and fix a full subcategory $A \subseteq \mathbf{pt}(\mathcal{E})$, where we identify points of $\mathcal{E}$ with their inverse images $\mathcal{E} \rightarrow \mathbf{Set}$. Generalizing the canonical vu-category structure, we can endow A with a canonical vf-category structure such that the evaluation map $\mathbf{ev} : \mathcal{E} \rightarrow [A, \mathbf{Set}]$ factors through $\mathbf{Sh}(A) := \mathbf{vfCat}(A, \mathbf{Set})$. **UT:** i personally don't like the notation $\mathbf{Sh}(\mathbf{pt}(\mathcal{E})_{|A})$ that much; if anything, i'd just write $\mathbf{Sh}(A)$ **GS:** i do agree, i changed it! Concretely, a vf-arrow $x \multimap_{s:\sigma} y_s$ in A is given by a natural transformation $x \Rightarrow \int_{s:\sigma} y_s$, where $\int_{s:\sigma} y_s$ is the *reduced product* functor.

We now show that vu-categories arising as points of a topos are taut.

Lemma 1.16. *Let $\phi : \tau \rightarrow \sigma$ in $\mathbb{U}$ and $(A_s)_{s:\sigma}$ a σ -family of sets. Then the image of $\int_{s:\sigma} A_s \hookrightarrow \int_{t:\tau} A_{\phi t}$ is described by the x such that $\psi_1^* x = \psi_2^* x$ for all pair of maps $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ equalizing ϕ .*

Proof. An element $\int_{t:\tau} A_{\phi t}$ that can be lift to $\int_{s:\sigma} A_s$ clearly satisfies the condition. We suppose now that $(a_t) \in \int_{t:\tau} A_{\phi t}$ cannot be lift to $\int_{s:\sigma} A_s$. We chose underlying sets $\phi : T \rightarrow S$ and we claim than the subsets $T_0 \times_{\phi(T_0)} T_0 \subseteq T \times_S T$, for T_0 τ -large, together with the subset $D = \{(t, t') \in T \times_S T \mid a_t \neq a_{t'}\}$ can be extended to a filter. We can thus extend them into ultrafilter λ on $T \times_S T$, the sets $(T_0 \times_{\phi(T_0)} T_0)$ ensure that the projections give maps $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ equalizing ϕ and the set D ensures that $\psi_1^*(a_t) \neq \psi_2^*(a_t)$. $\square$

Proposition 1.17. *Any sober vu-category $\mathcal{X} = \mathbf{pt}(\mathcal{E})$ is taut.*

Proof. We take some points $p, (q_s)_{s:\sigma}$ of $\mathcal{E}$ and $f : p \multimap_\tau q_{\phi t}$ in $\mathbf{pt}(\mathcal{E})$, that is, maps $f_E : E_p \rightarrow \int_{t:\tau} E_{q_{\phi t}}$ natural in $E \in \mathcal{E}$, that can be unthicken along $\phi : \tau \rightarrow \sigma$. It is enough to show that for any $\xi \in E_p$, $f_E(\xi)$ is in the image of $\int_{s:\sigma} E_{q_s} \hookrightarrow \int_{t:\tau} E_{q_{\phi t}}$. The hypothesis that f can be unthicken along ϕ ensures that $\psi_1^* f_E(\xi) = \psi_2^* f_E(\xi)$ for all $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ equalizing ϕ , and so we can conclude with Lemma 1.16. $\square$

Lemma 1.18. *The canonical vf-category structure on $A \subseteq \mathbf{pt}(\mathcal{E})$ is taut. **UT:** is it always? **GS:** i hope so! i will write the proof*

Proof. We first show that $\mathbf{Set}$ is taut. Let A and $(B_s)_{s:\sigma}$ be sets, we want to show that

$$\mathbf{Set} \left(A, \int_{t:\tau} B_{\phi t} \right)$$

is a sheaf over $(\tau \xrightarrow{\phi} \sigma) \in \mathbb{F}/\sigma$ for J_∞ . As $\mathbf{Set}(A, -)$ is continuous, it is enough to show that

$$\int_{t:\tau} B_{\phi t}$$

is a sheaf over $(\tau \xrightarrow{\phi} \sigma) \in \mathbb{F}/\sigma$.

GS: would be nice to fix on a notation for $(B, \{B\})$, i propose δ_B but not sure about it

First, in the case $\sigma = \star$, we noticed that $B^\tau = \mathbb{F}(\tau, (B, \{B\}))$, so it follows from J_∞ being sub-canonical.

In the general case, we denote $\pi : \sum_{s:\sigma} (B_s, \{B_s\}) \rightarrow \sigma$ in $\mathbb{F}/\sigma$ so that $\int_{t:\tau} B_{\phi t} = \mathbb{F}/\sigma(\phi, \pi)$, and so this follows from J_∞ on $\mathbb{F}/\sigma$ being subcanonical.

Now we show that $\mathbf{pt}(\mathcal{E})$ is taut [GS: à continuer](#)

□

The goal of the following is to prove that the functor $\mathbf{ev}: \mathcal{E} \rightarrow \mathbf{Sh}(A)$ is an equivalence when A is a separating set of points of $\mathcal{E}$.

We will obtain this by combining two more general results. Note that any vf-functor $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$ naturally induces a functor $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$ where, for $e \in \mathcal{E}$ [GS: i usually prefer capital letters for objects in a topos \(as their are thought as sheaf\)](#), the vf-functor $f^*(e): \mathcal{X} \rightarrow \mathbf{Set}$ is given:

- on objects, by mapping $x \in \mathcal{X}_0$ to the set $F(x)(e)$;
- on vf-arrows, by mapping $r: x \multimap_{s:\sigma} y_s$ to the component at e of the natural transformation $F(r): x \Rightarrow \int_{s:\sigma} y_s$.

Thus, we will now study how properties of F induce properties on f^* , in such a way that taking F to be the inclusion $A \hookrightarrow \mathbf{pt}(\mathcal{E})$ the corresponding functor $\mathcal{E} \rightarrow \mathbf{Sh}(A)$ will be an equivalence. Intuitively, we will proceed as if f^* were the inverse image of a geometric morphism $\mathbf{Sh}(\mathcal{X}) \rightarrow \mathcal{E}$, determining sufficient conditions on F making it localic and conditions making it hyperconnected. Since a geometric morphism which is both localic and hyperconnected is an equivalence, we will derive our conclusion in the case of $\mathcal{E} \rightarrow \mathbf{Sh}(A)$.

[UT: do we need to assume that \$\mathbf{Sh}\(\mathcal{X}\)\$ is a topos, or does infinitary-pretopos suffice? in principle we should have the latter with no problems](#)[GS: I hope we will only need infinitary-pretopos, i guess one can show the accessibility directly, but in our proofs it is deduced from the reconstruction theorem](#)

[UT: i assume the next theorems will assume tautness of \$\mathcal{X}\$, or at least make use of tautness of \$\mathbf{pt}\(\mathcal{E}\)\$?](#)[GS: I am actually not sure about how much tautness is used in these theorem, we need to check carefully.](#)

Theorem 1.19 (Localicness). *If $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$ is vf-faithful, then any vf-sheaf on $\mathcal{X}$ is a subquotient of a vf-sheaf in the image of $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$.*

Theorem 1.20 (Sub-hyperconnectedness). *If $F: \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$ is vf-full, then for each $e \in \mathcal{E}$ the frame homomorphism $\mathbf{Sub}_{\mathcal{E}}(e) \rightarrow \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*(e))$ induced by $f^*: \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$ is a quotient map.*

2. POINTS OF THE LOCALIC REFLECTION (5.3 OF SUJ)

We denote by $\rho_L: \mathbf{Topos} \rightarrow \mathbf{Loc}$ the usual localic reflection, i.e. $\rho_L(\mathcal{E}) := \mathbf{Sub}_{\mathcal{E}}(1)$.

Let $\mathcal{E}$ be a topos. Note that there is an evident vf-functor $\mathbf{pt}(\mathcal{E}) \rightarrow \iota_T(\mathbf{pt}(\rho_L(\mathcal{E})))$ defined on objects by mapping $p \in \mathbf{pt}(\mathcal{E})$ to its restriction $p|_{\mathbf{Sub}_{\mathcal{E}}(1)}: \mathbf{Sub}_{\mathcal{E}}(1) \rightarrow \mathcal{P}(1)$, which is a frame homomorphism and hence a point of $\rho_L(\mathcal{E})$. By the adjunction $\rho_T \dashv \iota_T$, this yields a continuous map $\Theta: \rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$.

Identified with its ι_T -image, the vf-functor Θ is bijective on objects and vf-faithful; we will now show that it is also vf-full, and hence an equivalence of vf-categories.

Lemma 2.1. *Let $\mathcal{C}$ be a small filtered category and $A: \mathcal{C} \rightarrow \mathbf{Set}$ such that there exist a family $(a_x)_{x:\mathbf{Ob}(\mathcal{C})}$. Then, there exists an inhabited filter λ such that the reduced product functor $A^\lambda: \mathcal{C} \rightarrow \mathbf{Set}$ defined by $x \mapsto A(x)^\lambda$ admits a global section.*

Proof. We consider the filter λ on $L := \prod_{x:\mathbf{Ob}(\mathcal{C})} \mathbf{Ob}(\mathcal{C}/x)$ defined by: $L_0 \subseteq L$ is λ -large iff there exists a non-empty finite sub-category $C \subseteq \mathcal{C}$, such that L_0 contains all the $(\theta^x: d^x \rightarrow x)_{x:\mathbf{Ob}(\mathcal{C})} \in L$ with the $(\theta^x)_{x:C}$ inducing a cone on C .

It is clear that λ defines a filter, and λ proper is equivalent to $\mathcal{C}$ being filtered, that is true by hypothesis.

We define a global section of A^λ given on $x \in \mathcal{C}$ by

$$(\theta_*^x(a_{d^x}))_{(\theta^x : d^x \rightarrow x) : \lambda} \in A(x)^\lambda.$$

We check the functoriality of this family along $f : y \rightarrow x$. This amounts to show that

$$(\forall(\theta^x : d^x \rightarrow x) : \lambda) f_*((\theta_*^y(a_{d^y}))) = \theta_*^x(a_{d^x})$$

and this is true by construction of λ (as for $(\theta^x : d^x \rightarrow x) : \lambda$ we can assume $f \circ \theta^y = \theta^x$). $\square$

Proposition 2.2. *The functor $\rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$ is vf-full, i.e., for any p and $(q_s)_{s:\sigma}$ such that $p \preceq_\sigma q_s$ in $\mathbf{pt}(\rho_L(\mathcal{E}))$, there exists $p \dashv_{\lambda \cdot \sigma} q_s$ in $\mathbf{pt}(\mathcal{E})$ for some filter λ inhabited.*

Proof. We take $p, (q_s) \in \mathbf{pt}(\mathcal{E})$ such that $p \preceq_\sigma^{\mathbf{pt}(\rho_L(\mathcal{E}))} q_s$ and we want to construct some $p \dashv_{\lambda \cdot \sigma} q_s$ in $\mathbf{pt}(\mathcal{E})$ for some proper filter λ .

We fix a small dense full subcategory $\mathcal{G} \subseteq \mathcal{E}$ of the topos $\mathcal{E}$ such that the category $\mathcal{C}$ of elements of the functor $p^* : \mathcal{G} \rightarrow \mathbf{Set}$ is filtered, for example given by a lex site.

Classically (apparently choice here!), $p \preceq_\sigma^{\mathbf{pt}(\rho_L(\mathcal{E}))} (q_s)$ implies that the functor

$$(E, \xi \in E_p) \in \mathcal{C} \mapsto \int_{s:\sigma} E_{q_s} \in \mathbf{Set}$$

satisfies the hypothesis of the lemma above. We obtain some filter λ and some family $(\alpha_E(\xi) \in \int_{\lambda \cdot \sigma} E_{q_s})$ natural in $(E, \xi \in E_p) \in \mathcal{C}$. This gives a natural transformation $\alpha \in \mathbf{Nat}_{E:S}(E_p, \int_{\lambda \cdot \sigma} E_{q_s})$ that we then can extend to all $\mathcal{E}$ by density of $\mathcal{G}$, obtaining thus some vu-arrow $p \dashv_{\lambda \cdot \sigma} (q_s)$ in $\mathbf{pt}(\mathcal{E})$. $\square$

Corollary 2.3. *The functor $\rho_T(\mathbf{pt}(\mathcal{E})) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}))$ is an equivalence of topological spaces.*

Hence, both ways to take get a topological space from a topos $\mathcal{E}$ —the points of the localic reflection or the topological reflection of its vf-category of points—coincide.

3. PROVING THE THEOREMS

We can deduce

Theorem 3.1 (5.4 SUJ). *If $F : \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$ is vf-full, then $f^* : \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$ is ‘algebraically subhyperconnect’, i.e., for any $X \in \mathcal{E}$ the map $\mathbf{Sub}_{\mathcal{E}}(X) \rightarrow \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*X)$ is a quotient of frames (that is, an embedding of locales).*

Proof. We consider the pullback

$$\begin{array}{ccc} \mathbf{Et}(f^*X) & \longrightarrow & \mathbf{pt}(\mathcal{E}/X) \cong \mathbf{Et}(X) \\ \text{ét} \downarrow & \lrcorner & \downarrow \text{ét} \\ \mathcal{X} & \xrightarrow{F} & \mathbf{pt}(\mathcal{E}) \end{array}$$

the above map $\mathbf{Et}(f^*X) \rightarrow \mathbf{pt}(\mathcal{E}/X)$ is still vf-full (vf-fullness is stable by pullback), and that its topological reflection $\rho_T(\mathbf{Et}(f^*X)) \rightarrow \rho_T(\mathbf{pt}(\mathcal{E}/X))$ also (vf-fullness is stable by topological reflection). Then by the above lemma we get

$$\rho_T(\mathbf{Et}(f^*X)) \rightarrow \mathbf{pt}(\rho_L(\mathcal{E}/X))$$

is vf-full. But a vf-full functor between vf-posets induces a quotient on the frames of opens, so

$$\mathcal{O}(\mathbf{pt}(\rho_L(\mathcal{E}/X))) \rightarrow \mathcal{O}(\rho_T(\mathbf{Et}(f^*X))) \cong \mathbf{Sub}_{\mathbf{Sh}(\mathcal{X})}(f^*X)$$

is surjective. It remains to notice that $\mathbf{Sub}_{\mathcal{E}}(X) \rightarrow \mathcal{O}(\mathbf{pt}(\rho_L(\mathcal{E}/X)))$ is surjective, and this follows from the more general fact that $\mathbf{Sub}(L) \rightarrow \mathbf{Sub}(\mathcal{O}(\mathbf{pt}(L)))$ is surjective, i.e., the restriction of a local to its points is a sublocal. $\square$

Corollary 3.2. *The functor $\mathcal{E} \rightarrow \mathbf{Sh}(\mathbf{pt}(\mathcal{E}))$ induces equivalences on subobjects.*

Proof. Its surjective on subobjects by the above. Using that $\mathcal{E}$ has enough points we conclude it is injective. $\square$

Now we can show the second part.

Theorem 3.3. *If $F : \mathcal{X} \rightarrow \mathbf{pt}(\mathcal{E})$ is faithful, then $f^* : \mathcal{E} \rightarrow \mathbf{Sh}(\mathcal{X})$ is ‘algebraically localic’, i.e., any $A \in \mathbf{Sh}(\mathcal{X})$ is a subquotient of a vf-sheaf f^*X for some $X \in \mathcal{E}$.*

Lemma 3.4. *Let $\pi : \mathcal{A} \rightarrow \mathcal{X}$ étale with $\pi : a \mapsto p$, then the following are equivalent:*

- (i) *for any two parallel vf-arrows $f, g : p \multimap (-)$, $f_*a = g_*a$.*
- (ii) *for any vf-arrow $f : p \multimap_{\sigma} q_s$ with lift $\tilde{f} : a \multimap_{\sigma} b_s$, then $F(q)(\sigma) = F(b)(\sigma)$*
- (iii) *if $(b_s)_{\sigma}$ converges to a in the topological reflection, then π is $b_*\sigma$ -injective.*
- (iv) *there exists an open $a \in V \subseteq \mathcal{A}$ such that $V \rightarrow \mathcal{X}$ is an embedding.*

SUJ 4.7.(i $\Rightarrow$ ii) For this we use the fact that the functor $\mathbf{opF} \rightarrow \mathbf{Set}$ defined by $\sigma \mapsto X^{\sigma}$, for each set X , maps kernel pairs to coequalizers. In practice, we might just make this follow from the fact that it is a J_{∞} -sheaf, which is needed to prove that $\mathbf{Set}$ is taut anyway.

Then the point is, lift $(q_s)_{s:\sigma} = \phi^*(r_l)_{l:\lambda}$, where $\phi : \sigma \twoheadrightarrow \lambda$ is the image factorization of $q : \sigma \rightarrow \mathcal{X}_0$, to $(b_s)_{s:\sigma}$. We need to express (b_s) as $\phi^*(?)$ too, and for this we use the sheaf condition to find a sequence $\lambda \rightarrow \mathcal{A}_0$. Then, diagram chase to prove that the two image factorizations coincide.

- (ii $\Rightarrow$ iii) Take the vf-arrow given by the convergence, see it as the unique lift of its projection, and apply (2).
- (iii $\Rightarrow$ iv) Apply (iii) to the filter of open neighborhoods of a , recalling that for topological spaces we can characterize convergence in terms of opens.
- (iv $\Rightarrow$ i) clear.

$\square$

proof of the theorem – SUJ 5.5-5.6. Let $A \in \mathbf{Sh}(\mathcal{X})$, it is enough to show that for any $a \in A_p$ for $p \in \mathcal{X}$ there is $X \in \mathcal{E}$, $V \subseteq f^*(X)$, and a map $V \rightarrow A$ touching a . It is enough to show that we can find a $X \in \mathcal{E}$ and $\xi \in X_{Fp}$ such that

$$\forall f, g : p \multimap_{\sigma} q_s \quad F(f)_*\xi = F(g)_*\xi \Rightarrow f_*a = g_*a.$$

as then it is easy to conclude by applying the above on the map $(a, \xi) \in \mathcal{A} \times_{\mathcal{X}} \mathbf{Et}(f^*X) \rightarrow \mathbf{Et}(f^*X)$. We now show the existence of such a $\xi \in X_{Fp}$.

On prend S un (petit) ensemble de $s = (f_s, g_s)$ couple de τ_s -flèches parallèles de domaine p , tq pour toute paire (f, g) de λ -flèches parallèles de domaine p il existe épaississement $\phi : \lambda \twoheadrightarrow \tau$ et $(f', g') \in S$ tels que $f = \phi^*f'$ et $g = \phi^*g'$.

On définit un filtre σ sur S . Pour tout $\xi \in X_{Fp}$ on note $A_{\xi} = \{(f, g) \in S \mid F(f)_*\xi = F(g)_*\xi\}$, et pour $\xi \in X_{Fp}$ et $\zeta \in Y_{Fp}$ on a $(\xi, \zeta) \in (X \times Y)_{Fp}$, et $A_{\xi} \cap A_{\zeta} = A_{(\xi, \zeta)}$. Donc les (A_{ξ}) définissent un filtre σ sur S (une partie de S est σ -large ssi elle contient un A_{ξ}).

On pose alors $f = (f_s)_{s:\sigma}$ et $g = (g_s)_{s:\sigma}$, ce sont deux $(s : \sigma) \cdot \tau_s$ -flèches. De plus, par définition de σ , pour tout $\xi \in X_p$, $F(f)_*\xi = F(g)_*\xi$, donc $F(f) = F(g)$, et par fidélité de F on déduit que $f = g$.

Ainsi, $f_*a = g_*a$, et donc $(f_s)_*a = (g_s)_*a$ pour tout $s : \sigma$, i.e., il existe un $\xi \in X$ tel que pour tout $s \in A_{\xi}$, $(f_s)_*a = (g_s)_*a$, i.e., tel que pour tout $(f, g) \in S$, si $F(f)_*\xi = F(g)_*\xi$ alors $f_*a = g_*a$.

Maintenant, pour (f, g) deux λ -flèches parallèles sortant de p quelconque, on peut par construction de S trouver un $\phi : \lambda \rightarrow \tau$ couvrant et $(f', g') \in S$ tels que $f = \phi^* f'$ et $g = \phi^* g'$. Alors si $F(f)_* \xi = F(g)_* \xi$, $F(\phi^* f')_* \xi = F(\phi^* g')_* \xi$, et $\phi^* F(f')_* \xi = \phi^* F(g')_* \xi$, mais comme ϕ est couvrant, ϕ^* est injectif et donc $F(f')_* \xi = F(g')_* \xi$, d'où $f'_* a = g'_* a$ et $\phi^* f'_* a = \phi^* g'_* a$ soit $f_* a = g_* a$. $\square$

Corollary 3.5. *The functor $\mathcal{E} \rightarrow \text{Sh}(\text{pt}(\mathcal{E}))$ is an equivalence of categories.*

Proof. todo $\square$

APPENDIX A. TAUTNESS: THE SHEAF DEFINITION

We first begin with the most concrete and most straightforward definition of tautness.

First notice that if $f = \phi^*g : p \multimap_\tau (q_{\phi t})_{t:\tau}$ is the thickening of some $g : p \multimap_\sigma (q_s)$, then for any $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ equalizing ϕ (i.e., $\phi \circ \psi_1 = \phi \circ \psi_2$) we have $\psi_1^*f = \psi_2^*f$.

A vu-category $\mathcal{X}$ is said *taut* if the reverse is true, we can always unthicken a vu-arrow if we can (and if so, in a unique way):

Definition A.1. For any $f : p \multimap_\tau (q_{\phi t})_{t:\tau}$, such that

$$(\forall \psi_1, \psi_2 : \lambda \rightrightarrows \tau \text{ equalizing } \phi) \psi_1^*f = \psi_2^*f$$

(we will say f is unthickenable along ϕ if its the case)

there exists a unique $g : p \multimap_\sigma (q_s)$ such that $f = \phi^*g$.

This definition can be understand as a ‘internal (Cauchy-)completeness of the objects of $\mathcal{X}$ ’: in the case $\mathcal{X}$ being the ultracategory of complete metric spaces, it really gives that any Cauchy-ultrasequence converge.

Definition A.2. Another way to express the definition above is to ask the following functor $(\mathbb{U}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$ whose functoriality encodes the thickening operation

$$(\phi : \tau \rightarrow \sigma) \mapsto \mathcal{X}(p, (q_{\phi t})_{t:\tau})$$

to be a sheaf for the atomic topology $\mathbf{Sh}(\mathbb{U}_{at})$.

This definition can thus be understood as ‘the thickening is a sheaf for the topology on $\mathbb{U}$ ’,

Remark A.3. In light of Eliasson result that $\mathbf{Sh}(\mathbb{U}_{at}) = \mathbf{Sh}(\mathbb{F}, J_\infty)$ we expect that, under tautness assumption, we can identify vu and vf categories. Thus suggesting that we get a notion of 1-space that is no more dependent of the ultrafilter calculus.

APPENDIX B. VU-VF?

Let $\sigma \in \mathcal{F}$ be a filter, we denote $\mathbf{Ext}(\sigma) := \{\dot{\sigma} \hookrightarrow \sigma\}$ the set of the subobjects of σ in $\mathcal{F}$ that are ultrafilters. Moreover $\mathbf{Ext}(\sigma) \subseteq \beta(S)$ has a compact Hausdorff topology (and this topology doesn’t depend of the choice of S), for $(\dot{\sigma}_\ell)_{\ell:\lambda}$ in $\mathbf{Ext}(\sigma)$, we denote $\int_{\ell:\lambda} \dot{\sigma}_\ell \in \mathbf{Ext}(\sigma)$ its limit in $\mathbf{Ext}(\sigma)$.

Remark B.1. For a choice of underlying set S , we can identify $\mathbf{Ext}(\sigma)$ with the set of ultrafilters $\dot{\sigma} \in \beta(S)$ extending σ . Moreover $(\dot{\sigma} \hookrightarrow \sigma)_{\dot{\sigma} \in \mathbf{Ext}(\sigma)}$ is jointly epic in $\mathbb{F}$, and so is a cover for J_∞ .

Moreover, the category $\mathbb{U}/\sigma$ is isomorphic to the disjoint union $\bigsqcup_{\dot{\sigma} \in \mathbf{Ext}(\sigma)} \mathbb{U}/\dot{\sigma}$.

Construction B.2. Let $\mathcal{X}$ be an vu-category, we want to define a vf-category $\mathbf{vf}(\mathcal{X})$. The objects are the same. For $p, (q_s)_{s:\sigma}$ and $\sigma \in \mathbb{F}$, we define

$$\mathbf{vf}(\mathcal{X})(p, (q_s)_{s:\sigma}) \subseteq \prod_{\dot{\sigma} \in \mathbf{Ext}(\sigma)} \mathcal{X}(p, (q_s)_{s:\dot{\sigma}})$$

to be the subset consisting of families $(f_{\dot{\sigma}} : p \multimap_{\dot{\sigma}} q_s)$ satisfying the following *compatibility condition*:

for any (injective) family $(\dot{\sigma}_\ell \in \mathbf{Ext}(\sigma))_{\ell:\lambda}$, the vu-arrow

$$(f_{\dot{\sigma}_\ell}) \circ \delta : p \multimap_\lambda (p \multimap_{\dot{\sigma}_\ell} q_{t\ell})$$

is equal the thickening of $f_{\int_{\ell:\lambda} \dot{\sigma}_\ell}$ along $\sum_{\ell:\lambda} \dot{\sigma}_\ell \rightarrow \int_{\ell:\lambda} \dot{\sigma}_\ell$.

Remark B.3. As $\mathbf{Ext}(\sigma) = \sigma$ if σ is an ultrafilter, $\mathbf{vf}(\mathcal{X})$ has the same vu-arrows that $\mathcal{X}$.

Construction B.4. We now to define the composition. Let $f : p \multimap_{\sigma} q_s$ in $\mathbf{vf}(\mathcal{X})(p, (q_s)_{s:\sigma})$ given by $f = (f_{\dot{\sigma}})_{\dot{\sigma}:\text{Ext}(\sigma)}$, and a σ -family of $(g^s : q_s \multimap_{t:\tau_s} r^s_t)$ in $\mathbf{vf}(\mathcal{X})(q_s, (r^s_t)_{t:\tau_s})$ given by $g^s = (g^s_{\dot{\tau}_s})_{\dot{\tau}_s:\text{Ext}(\tau_s)}$.

To define the composite, we need for each $\theta \in \text{Ext}(\sum_{s:\sigma} \tau_s)$ some

$$h_{\theta} : p \multimap_{(s,t):\lambda} r^s_t$$

for any $\dot{\sigma} \in \text{Ext}(\sigma)$ and $(\dot{\tau}_s \in \text{Ext}(\tau_s))_{s:\dot{\sigma}}$, we let

$$h_{\sum_{\dot{\sigma}} \dot{\tau}_s} = (g^s_{\dot{\tau}_s}) \circ f_{\dot{\sigma}}$$

prescribing the value of h on the subset

$$\sum_{\dot{\sigma} \in \text{Ext}(\sigma), (\dot{\tau}_s \in \text{Ext}(\tau_s))} \int_{s:\sigma} \text{Ext}(\dot{\tau}_s) \subseteq \text{Ext} \left(\sum_{s:\sigma} \tau_s \right)$$

that is dense (any point in the right space is a limit point of an ultrafamily from the left space).

Hence the value of h is determined everywhere, we need to show the compatibility condition for it...

Lemma B.5. Let λ be an ultrafilter, and $\sigma_{\ell} \in \text{Ext}(F)$ converging to $\int_{\ell:\lambda} \sigma_{\ell} = \sigma \in \text{Ext}(F)$, and for $s : \sigma_{\ell}$ an ultrafilter $\tau_{\ell}^s \in \text{Ext}(G_s)$ converging to $\int_{\ell:\lambda} \tau_{\ell}^s = \tau^s \in \text{Ext}(G_s)$, , then

$$(h_{\sum_{s:\sigma_{\ell}} \tau_{\ell}^s}) \circ \delta : a \rightarrow (a)_{\ell:\lambda} \rightarrow (c_{s,t})_{(\ell:\lambda) \cdot (s:\sigma_{\ell}) \cdot (t:\tau_{\ell}^s)}$$

and

$$(h_{\sum_{s:\sigma} \tau^s}) : a \rightarrow (c_{s,t})_{(s:\sigma) \cdot (t:\tau^s)}$$

coincide (modulo thickening).

APPENDIX C. DIRECT IMAGE

We try to give an explicit description of the direct image.

We take $F : \mathcal{X} \rightarrow \mathcal{Y}$ a vf-functor, some $A \in \mathbf{Sh}(\mathcal{X})$, and want to define $B = f_*(A) \in \mathbf{Sh}(\mathcal{Y})$ the direct image (this always exists by the special adjoint theorem if A is bounded) that can be categorically thought as a right Kan-extension.

We define $B(y)$ as the data of families $(\xi^h_s \in A(x_s))_{s:\sigma}$ for all $h : y \multimap_{\sigma} F(x_s)$, such that

- (i) if $h : y \multimap_{\sigma} F(x_s)$ and $(f_s : x_s \rightarrow x'_s)$, then $\int A(x_s) \rightarrow \int A(x'_s)$ sends ξ^h to $\xi^{(f_s) \circ h}$.
- (ii) if $h : y \multimap_{\sigma} F(x_s)$ and $\phi : \tau \rightarrow \sigma$, then $\int_{\sigma} A(x_s) \rightarrow \int_{\tau} A(x_{\phi t})$ sends ξ^h to $\phi^*(\xi^h)$.
- (iii) if $(h^{\ell} : y \multimap_{\sigma_{\ell}} F(x^{\ell}_s))_{\ell:\lambda}$ merge into $h : y \multimap_{\lambda} (y \multimap_{\sigma_{\ell}} F(x^{\ell}_s))$ then $\xi^h \in \int_{\ell:\lambda} \int_{s:\sigma_{\ell}} A(x^{\ell}_s)$ equals $(\xi^{h^{\ell}} \in \int_{s:\sigma} A(x^{\ell}_s))_{\ell:\lambda}$

This can be seen as taking the rep-vf-category of vf-arrows $y \multimap_{-} F(-)$, given a vf-sheaf on it given by $\int_{-} A(-)$, and saying $B(y)$ is the global element of this sheaf.

We need to show this is small, and functorial. The smallness can follow from assuming $\mathcal{X}$ and $\mathcal{Y}$ small 1-spaces (but we can surely ask less). Let's try to give the functoriality, we take $g : y \multimap_{\tau} y'_t$ in $\mathcal{Y}$, and some $(\xi^h)_h$ defining an element of $B(y)$, we want an element of $\int_{t:\tau} B(y'_t)$.

We have an inclusion (where the limits are taking with the condition above, maybe not exactly limits...)

$$\int_{t:\tau} B(y'_t) = \int_{t:\tau} \lim_{h:y'_t \multimap_{\sigma_t} F(x^t_s)} \int_{s:\sigma_t} A(x^t_s) \hookrightarrow \lim_{(h_t:y'_t \multimap_{\sigma_t} F(x^t_s))} \int_{t:\tau} \int_{s:\sigma_t} A(x^t_s)$$

The $(\xi^h)_h$ gives for any $(h_t : y'_t \multimap_{\sigma_t} F(x^t_s))_{t:\tau}$ an element $(\xi^{(h_t) \circ g})$ of $\int_{\tau} \int_{\sigma_t} A(x^t_s)$, so an element of the right set. We need to show it is in the left one.

APPENDIX D. TAUTNESS 2: THE STACK DEFINITION

In the posetal case, small vu-poset are sober if and only if they are taut, we denote $X \rightarrow \rho_T(X)$ the tautification of a vu-poset. One can extend to the following

Definition D.1. A vu-category $\mathcal{X}$ if it is right orthogonal to all $X \rightarrow \rho_T(X)$ for (small) X vu-poset, i.e., precomposing with $X \rightarrow \rho_T(X)$ induces an equivalence of category

$$\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \xrightarrow{\sim} \mathbf{vuCat}(X, \mathcal{X}).$$

Lemma D.2. Let $f : p \multimap_\tau (q_{\phi t})$. We chose $\phi = T \rightarrow S$ representing $\tau \rightarrow \sigma$, and consider the vu-poset $X = \{x\} \sqcup S$ with only convergences $x \preceq_\lambda (\rho_\ell)$ for some $\rho \in S^\lambda$ with $\rho : \lambda \rightarrow \sigma$ iff ρ factorizes (not necessarily uniquely) through $\rho : \lambda \rightarrow \tau \rightarrow \sigma$.

Then f is unthickenable, iff the map $X \rightarrow \mathcal{X}$ mapping $x \preceq_\tau (t)$ to f , extends to a vu-functor.

Proof. We suppose it is unthickenable. We then have a functor $X \rightarrow \mathcal{X}$ sending $x \mapsto p$ and $s \mapsto q_s$, and $x \preceq_\lambda (\rho_\ell)$ is sent on $\rho'^* f : p \multimap_\lambda (q_{\rho t})$, (for any $\rho' : \lambda \rightarrow \tau$ factorizing ρ , it doesn't depend of the choice of it by hypothesis).

The other direction is similar. □

proof: equivalence with the sheaf definition. Let suppose the orthogonality condition, and $f : p \multimap_\tau (q_{\phi t})$ such that $\psi_1^* f = \phi_2^* f$ for all ϕ_i equalizing ϕ . And we take X as in the lemma above.

Then $\rho_T(X)$ is given by the topological space $X = \{x\} \sqcup S$ with only convergence $x \preceq_\sigma (s)$.

Hence, the orthogonality gives us a unique $g : p \multimap_\sigma (q_s)_{s:\sigma}$ unthickening f as wanted.

We suppose now the sheaf condition. The uniqueness of the unthickening gives that $\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \rightarrow \mathbf{vuCat}(X, \mathcal{X})$ is fully-faithful:

indeed, let $F, G : \rho_T(X) \rightarrow \mathcal{X}$ and a natural transformation on X , i.e., for any $x \in X$ a $\alpha_x : F(x) \rightarrow G(x)$ such that for any $f : x \preceq_\sigma (y_s)$ in X we have $G(f) \circ \alpha_x = (\alpha_{y_s}) \circ F(f)$. Then it is clear that this can be extended in at most one way to a natural transformation $\alpha : F \Rightarrow G$, and that this is the case iff the naturality condition holds for any f in the tautification of X . But if $f : x \preceq_\sigma (y_s)$ is in the tautification of X , there exists a ϕ such that $\phi^* f : x \preceq_\tau (y_{\phi t})$ is in X and so $G(\phi^* f) \circ \alpha_x = (\alpha_{y_{\phi t}}) \circ F(\phi^* f)$, that means $\phi^* G(f) \circ \alpha_x = (\alpha_{y_{\phi t}}) \circ \phi^* F(f)$, or again $\phi^*(G(f) \circ \alpha_x) = \phi^*((\alpha_{y_s}) \circ F(f))$, and we conclude $G(f) \circ \alpha_x = (\alpha_{y_s}) \circ F(f)$ by uniqueness of the unthickening.

We now show $\mathbf{vuCat}(\rho_T(X), \mathcal{X}) \rightarrow \mathbf{vuCat}(X, \mathcal{X})$ to be essentially surjective, we take some $F : X \rightarrow \mathcal{X}$ and we want to show it can be extended to some $G : \rho_T(X) \rightarrow \mathcal{X}$. On the objects G is F , we take now a vu-arrow $f : x \preceq_\sigma (y_s)$ in $\rho_T(X)$, i.e., such that $\phi^* f : x \preceq_\tau (y_{\phi t})$ is true in X . We can show that $F(\phi^* f)$ satisfies the condition to be unthicken (we can take the $X_0 = \{x\} \sqcup S$ from above and sent it to X and use the lemma above). Hence G is defined on vu-arrows, the functoriality come again by the functoriality of F by uniqueness of the unthickening. □

APPENDIX E. TAUTNESS 3: THE ALGEBRAIC DEFINITION

Let $\phi : \tau \rightarrow \nu$ and $f : p \multimap_\tau (q_{\phi t})$ unthickenable. The magic choice lemma on $\mathbf{U}$ allows us to 'make ϕ a projection': we can find λ and $\psi : \lambda \cdot \sigma \rightarrow \tau$ such that $\phi \circ \psi = \pi_1$. Then both maps $\lambda \cdot \lambda \cdot \sigma \rightrightarrows \lambda \cdot \sigma \rightarrow \tau$ equalize ϕ , and so $\psi^* f : p \multimap_{\lambda \cdot \sigma} (q_s)$ have the same thickening along both projections $\lambda \cdot \lambda \cdot \sigma$.

Hence we get the following

Proposition E.1. *For $\mathcal{X}$ to be taut, it is enough to ask for any $f : p \multimap_{\lambda \cdot \sigma} (q_s)$ such that $\pi_1^* f = \pi_2 * f : p \multimap_{\lambda \cdot \lambda \cdot \sigma} (q_s)$ to be of the form $f = !_{\lambda}^* g$ for some $g : p \multimap_{\sigma} (q_s)$.*

Or, abstracting the hom functor

Proposition E.2. *For $H : (\mathbb{U}/\sigma)^{\text{op}} \rightarrow \mathbf{Set}$ to be a sheaf, it is enough to have for any $\phi : \tau \rightarrow \sigma$ and $\lambda \in \mathbb{U}$, the coequalize in $\mathbb{U}$*

$$\lambda \cdot \lambda \cdot \tau \rightrightarrows \lambda \cdot \tau \rightarrow \tau$$

induces the coequalizer in $\mathbb{U}/\sigma$

$$\lambda \cdot \lambda \cdot \phi \rightrightarrows \lambda \cdot \phi \rightarrow \phi$$

induces a equalizer of sets

$$A(\lambda \cdot \lambda \cdot \phi) \rightrightarrows A(\lambda \cdot \phi) \rightarrow A(\phi)$$

It is not hard to check that this à priori stronger condition is satisfied on the sober vu-category. We will show it is actually not stronger.

On peut montrer? qu'un sheaf sur $\mathbb{U}/\sigma$ preserve les coeq de la forme

$$(k : \kappa) \cdot \lambda_k \cdot \lambda_k \cdot \tau_k \rightrightarrows (k : \kappa) \cdot \lambda_k \cdot \tau_s \rightarrow (k : \kappa) \cdot \tau_k$$

pour une certaine map $(k : \kappa) \cdot \tau_k \rightarrow \sigma$

Proposition E.3. *Let $H : \mathbb{U}^{\text{op}} \rightarrow \mathbf{Set}$ a sheaf, then*

$$H(\star) \rightarrow H(\tau) \rightrightarrows H(\tau \cdot \tau)$$

is an equalizer.

Proof. Let $\xi \in H(\tau)$ such that $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\tau \cdot \tau)$, for $\psi_1, \psi_2 : \lambda \rightrightarrows \tau$ we will show that $\psi_1^*(\xi) = \psi_2^*(\xi)$:

$$\begin{array}{ccc} \lambda \cdot \lambda & \xrightarrow{\pi_1} & \lambda \\ \psi_1 \cdot \psi_2 \downarrow & & \downarrow \psi_1 \downarrow \psi_2 \\ \tau \cdot \tau & \xrightarrow{\pi_2} & \tau \end{array}$$

As H sends any map to a mono, it is enough to show that $\pi_1^*(\psi_1^*(\xi)) = \pi_1^*(\psi_2^*(\xi))$

$$\pi_1^*(\psi_1^*(\xi)) = (\psi_1 \cdot \psi_2)^*(\pi_1^*(\xi)) = (\psi_1 \cdot \psi_2)^*(\pi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$$

Hence, for all $\psi_1, \psi_2 : \tau \rightrightarrows \sigma$ we have $\pi_1^*(\psi_1^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$

In particular (using $\psi_2 \cdot \psi_2$ instead of $\psi_1 \cdot \psi_2$) we get

$$\pi_1^*(\psi_2^*(\xi)) = (\psi_2 \cdot \psi_2)^*(\pi_1^*(\xi)) = (\psi_2 \cdot \psi_2)^*(\pi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$$

$\pi_1^*(\psi_2^*(\xi)) = \pi_2^*(\psi_2^*(\xi))$. and so $\pi_1^*(\psi_1^*(\xi)) = \pi_2^*(\psi_2^*(\xi)) = \pi_1^*(\psi_2^*(\xi))$ as wanted. $\square$

A first generalization

Proposition E.4. *Let $H : \mathbb{U}^{\text{op}} \rightarrow \mathbf{Set}$ a sheaf, then*

$$H(\kappa) \rightarrow H(\tau \cdot \kappa) \rightrightarrows H(\tau \cdot \tau \cdot \kappa)$$

is an equalizer.

Proof. Let $\xi \in H(\tau \cdot \kappa)$ such that $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\tau \cdot \tau \cdot \kappa)$, for equalizing

$$\psi_1, \psi_2 : \lambda \rightrightarrows \tau \cdot \kappa \rightarrow \kappa$$

we will show that $\psi_1^*(\xi) = \psi_2^*(\xi)$:

It is enough to give some

$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \tau \cdot \tau \cdot \kappa & \xrightarrow[\pi_2]{\pi_1} & \tau \cdot \kappa \xrightarrow{\pi_2} \kappa \end{array}$$

such that $\pi_2 f_1 = \pi_2 f_2$ and $\pi_1 f_1 = \psi_1 g$, $\pi_1 f_2 = \psi_2 g$ and we conclude as upside. $\square$

Proposition E.5. *Let $H : \mathbb{U}^{\text{op}} \rightarrow \mathbf{Set}$ a sheaf, then*

$$H(\sigma \cdot \kappa_s) \rightarrow H(\sigma \cdot \tau_s \cdot \kappa_s) \rightrightarrows H(\sigma \cdot \tau_s \cdot \tau_s \cdot \kappa_s)$$

is an equalizer.

Proof. Let $\xi \in H(\sigma \cdot \tau_s \cdot \kappa_s)$ such that $\pi_1^*(\xi) = \pi_2^*(\xi) \in H(\sigma \cdot \tau_s \cdot \tau_s \cdot \kappa_s)$, for equalizing

$$\psi_1, \psi_2 : \lambda \rightrightarrows \sigma \cdot \tau_s \cdot \kappa_s \rightarrow \sigma \cdot \kappa_s$$

we will show that $\psi_1^*(\xi) = \psi_2^*(\xi)$:

It is enough to give some

$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \sigma \cdot \tau_s \cdot \tau_s \cdot \kappa & \xrightarrow[\pi_2]{\pi_1} & \sigma \cdot \tau_s \cdot \kappa_s \xrightarrow{\pi_2} \sigma \cdot \kappa_s \end{array}$$

such that $\pi_2 f_1 = \pi_2 f_2$ and $\pi_1 f_1 = \psi_1 g$, $\pi_1 f_2 = \psi_2 g$ and we conclude as upside. $\square$

Question E.6. En générale, quelle condition il faut sur un coeq

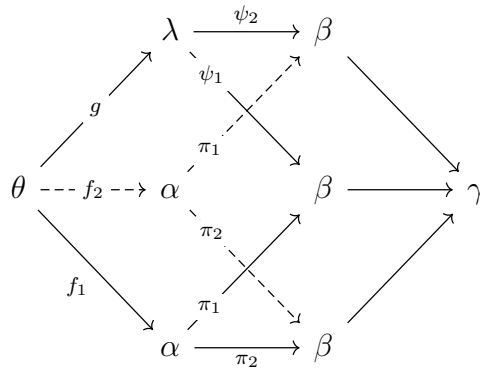
$$\alpha \rightrightarrows \beta \rightarrow \gamma$$

pour avoir:

pour tout $\psi_1, \psi_1 : \lambda \rightrightarrows \beta$ equalizing $\beta \rightarrow \gamma$ there exists such a diagram

$$\begin{array}{ccc} \theta & \xrightarrow{\quad g \quad} & \lambda \\ f_1 \downarrow \downarrow f_2 & & \psi_1 \downarrow \downarrow \psi_2 \\ \alpha & \xrightarrow[\pi_2]{\pi_1} & \beta \longrightarrow \gamma \end{array}$$

such that $\pi_2 f_1 = \pi_2 f_2$ and $\pi_1 f_1 = \psi_1 g$, $\pi_1 f_2 = \psi_2 g$.



In $\mathbb{U}$ we can complete span, but not in general cubes! (it is not filtered category as we cannot complete parallel arrow, a demi-cubes has equations in it contrary to a demi-square).

$$\alpha \times_{\pi_2=\pi_2} \alpha \longrightarrow \beta \times_{\gamma} \beta$$

We thus need to show that the following are covering

$$(\mathcal{T}.\mathcal{T}.\mathcal{K}) \times_{\star.\mathcal{T}.\mathcal{K}} (\mathcal{T}.\mathcal{T}.\mathcal{K}) \xrightarrow{\pi_1, \pi_1} (\mathcal{T}.\mathcal{K}) \times_{\mathcal{K}} (\mathcal{T}.\mathcal{K})$$

and these morphisms are respectively isomorphic to

$$(\tau \times \tau). \tau. \kappa \longrightarrow (\tau \times \tau). \kappa$$

APPENDIX F. FILTERS AS GERMS OF CLOSED, AN ALTERNATIVE

We give now a similar realization of filters, that differs from the one of [Moer]. A filter $\sigma \in \mathbb{F}$ on S , can be identified to the subset of the $\sigma \in \beta(S)$ that extend σ , this is a closed subset that we denoted $\text{Ext}(\sigma) \subseteq \beta(S)$. And we can do the same than above, replacing germ by open neighborhood of a closed subset.

This equivalent definition of $\mathbb{F}$ can be then given as follows (but its (à priori?) non-constructive! still an interesting intuition though). The objects are pairs of a set S with a closed subspace $K \subseteq \beta(S)$ (this corresponds bijectively to the filter whose $S_0 \subseteq S$ is large iff $K \subseteq \beta(S_0)$)

Proposition F.1. *These constructions are inverse from each other.*

Proof. From σ a filter on S , we get $K \subseteq \beta(S)$ of ultrafilter extending σ , we need to show that for $S_0 \subseteq S$, $S_0 \in \sigma$ iff $K \subseteq \beta(S_0)$: if $S_0 \in \sigma$ and $\sigma \in K$, i.e., σ extending σ , then S_0 is also σ -large and so $\beta(S_0) \ni \sigma$; conversely if $S_0 \notin \sigma$ then we can extend σ into an ultrafilter σ' containing $S \setminus S_0$ and so $\sigma' \in K$ but $\sigma' \notin \beta(S_0)$.

From $K \subseteq \beta(S)$ a closed subset, we get σ the filter of S_0 such that $K \subseteq \beta(S_0)$, and we need to show for $\sigma \in \beta(S)$, that $\sigma \in K$ iff it extends σ : if $\sigma \in K$ and $S_0 \in \sigma$, i.e., $K \subseteq \beta(S_0)$, then $\sigma \in \beta(S_0)$ and $S_0 \in \sigma$ as wanted; conversely if $\sigma \notin K$ then, as K is closed and the topology is generated by the $\beta(S_0)$, we can find S_0 such that $K \subseteq \beta(S_0)$ but $\sigma \notin \beta(S_0)$. $\square$

All of this uses ultrafilter... but we can go localic!

The local $\beta(S)$ is given by the poset $Fil(S)$ of filters on S ordered by usual inclusion (as above the open $U \subseteq \beta(S)$ corresponds to the filter of S_0 such that $\beta(S_0) \cup U = \beta(S)$), that is indeed a frame.

We can defined the category of germs of closed subset: objects are given by a locale L and an open $U \in \mathcal{O}(L)$, and a map from (L, U) to (L', U') is given by an open $V \in \mathcal{O}(L)$ such that $V \vee U = L$ and a map of locales $f : V \rightarrow L'$ such that $f^{-1}(U') \subseteq U \wedge V$, modulo the equivalence relation identifying two such (V, f) (V', f') if f and f' coincide when restricted to $V \wedge V'$.

This defined the category of germes of closed subsets. If we restricted to locales of the form $\beta(S)$ (or explicetely the frame given by the poset of filters on S) and ask f to be open maps, we get the category of filter. It would be interesting to see what happens if we relax the condition: considering all locales but still open maps, or all locals and all maps.

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